

Chalk's Ocean Airways 101

Flight model: Grumman G-73T Turbo Mallard
Date: December 19, 2005

From: Miami, US
To: Bimini, Bahamas

The Grumman G-73T Turbo Mallard is a twin propeller engine with a v share bottom that lets it to land in the waters.

Problem

The flight just minutes after the take off at an altitude of 500 meters the passengers all of the suddenly hears a loud bang followed by a heavy flame in left side of the plane. The plane then begins to bang towards the left and drives sharply. The pilots had little time to react and the plane crashed into the sea with no survivors.

Investigation

In the investigation, it was found that the only recorder in the plane was a CVR and the plane didn't have a FDR. From the footage captured by one of the witnesses it was proven that the left wing got ripped off the plane in the mid air. Also, The CVR recovered was found that the erase feature of the CVR was not working properly which made the voice being recorded on top of the previous other. The weather on the day of crash was found to be calm and normal thus ruling out the probability of bad weather that could have torn the plane apart. The investigators find a report by the FAA that cautions the company that a fault part in the propeller in the same aircraft could come out if not being replaced. But, the investigators find that the propellers were not separated during the flight but were found to be intact. The investigators notice **sooting** on the fractured edges of the damaged wing, the sooting is a phenomena which occurs when the wings are exposed to rapid fire. Most of the fractures that the investigators find is due to over stress of metal. In a plane wing, the **spars** run the length of the wing and in between these spars there are **stringers** that supports these spars. Comprise in the strength of any one of these would lead to loss of strength of the plane to carry the load. These were many frequent **corrosion repairs** in all the companies fleet of plane with poor workmanship. A **doubler** is used to patch the defective area acting as an second skin for the plane. The crack on the doubler of the broken part of the wing was found to have occurred due to metal fatigue. It was found that the mechanics have tried to stop the cracks from expanding further. This was done by a method called as a **stop drill hole**, where in a hole is made at the end of the crack which distributes the stress instead of concentrating them at one point. Despite the efforts the crack seems to have kept growing. Then a doubler was added on the both sides of the surface of the wing (Top and bottom of the wing) on the defective patch. Even then the crack seems to have continues to grow. The investigators find there was frequent fuel leaks in the left wing of the plane being reported. When the plane fuel tank was examined it was found that there was a crack in one of the stringers of the wing.

Changes that were brought after the incident

The report advises FAA had to inspect these small fleet of plane closely which they had failed to do prior when compared to the large carrier fleets which are monitored more vigorously.

Reference

1. Cracks In The System — Ocean Crashes — FULL EPISODE — Mayday: Air Disaster
<https://www.youtube.com/watch?v=-RR1X0T3Gps>